



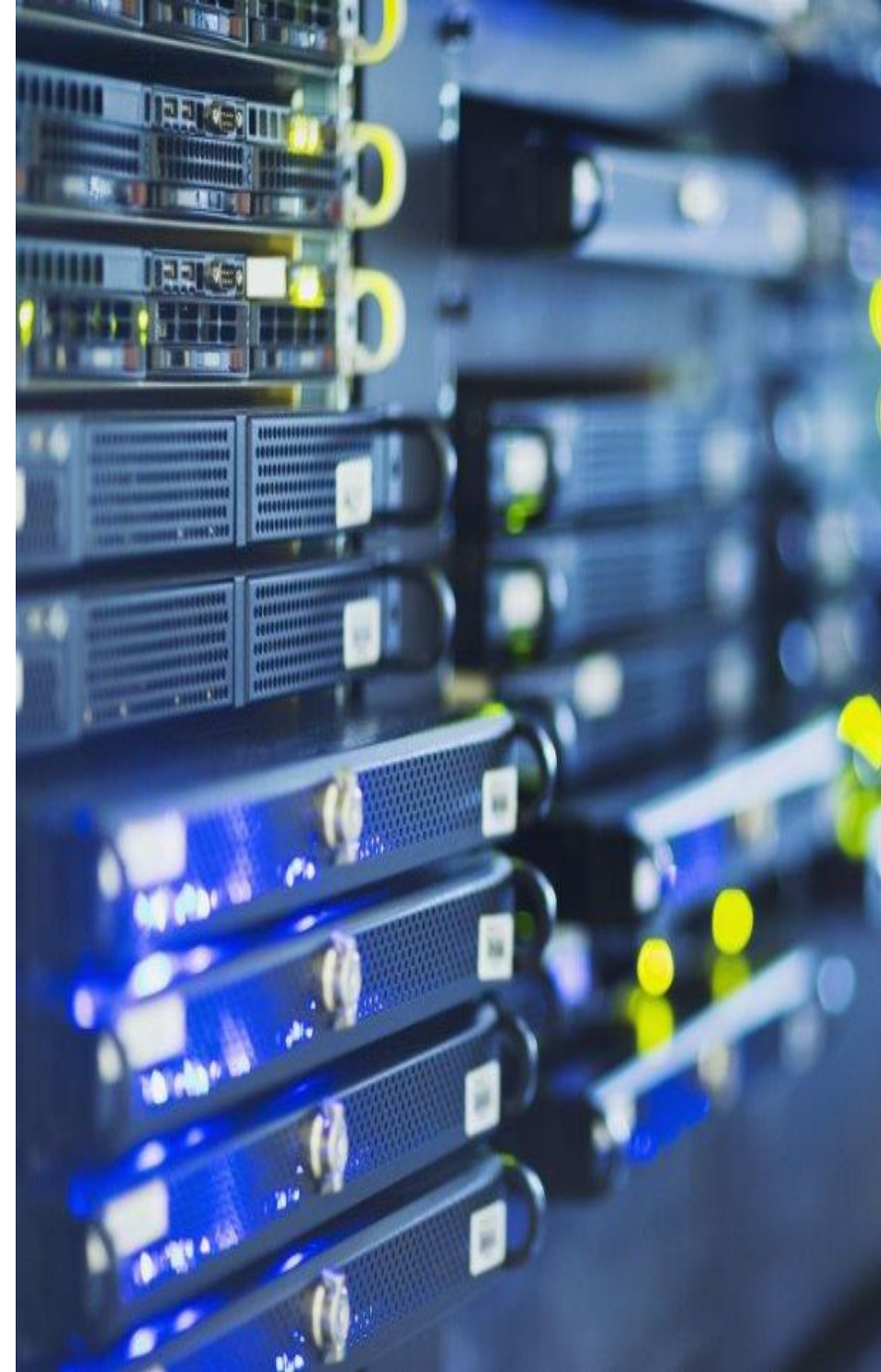
Cloud computing

Machines, environments and workloads



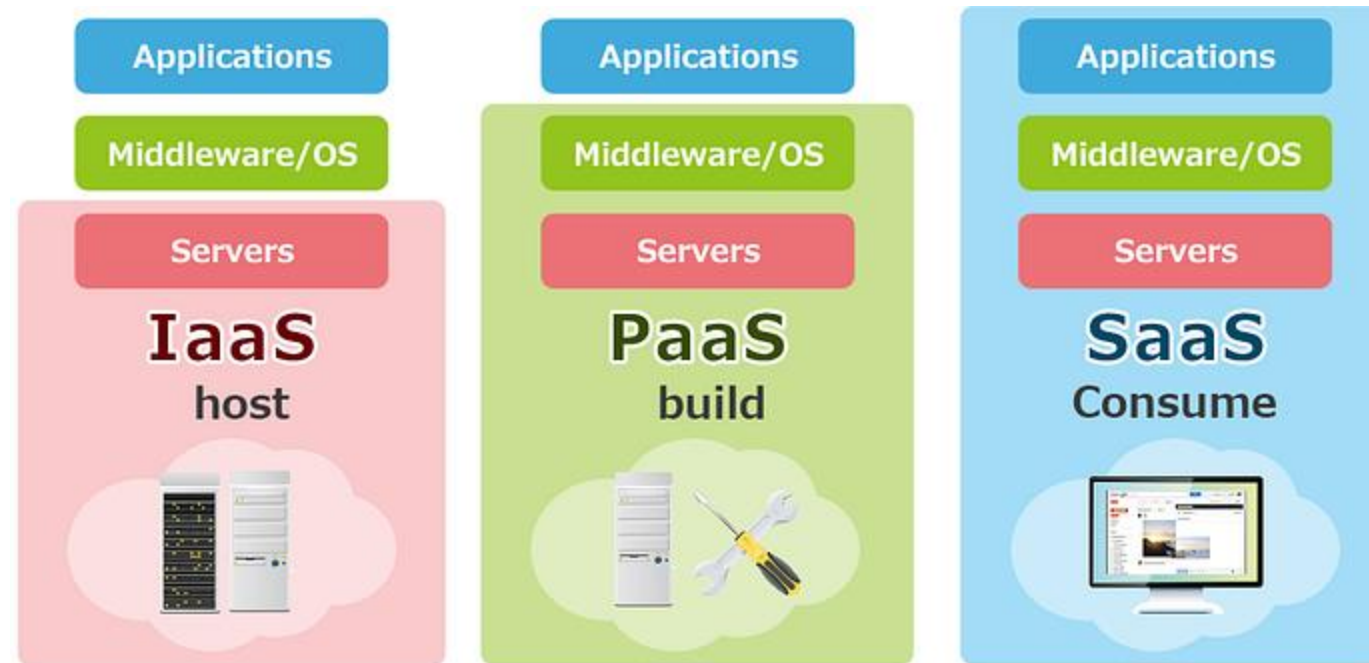
What is cloud computing?

- The process of supplying computing services over the internet
 - Servers, storage, databases, networking, software ...
- The user can reserve a specific amount of computing resources
 - Single computers or whole clusters
 - Computers with GPUs or without
 - High customizability
- Physically hosted as a cluster computer
 - The hardware is often similar to supercomputers



Different implementations of cloud services

- Three often-used categories: IaaS, PaaS and SaaS



Different implementations of cloud services: IaaS

- Infrastructure as a service (IaaS)
 - User reserves a number of resources (CPU's, RAM, memory)
 - High customizability
 - Often virtual machines on top of physical hardware in practice
 - e.g. cPouta and ePouta
- "Bare-metal" nodes
 - The user gets access to a given physical server
 - High customizability
 - A lot of resources given to a user which might end up being wasted (a whole server at a time)
 - e.g. Chameleon Cloud



Different implementations: PaaS

- Platform as a service (PaaS)
 - All basic infrastructure already set up
 - Includes networks, OS, compilers, etc.
 - Easy way for a user to start developing in a new environment
 - e.g. Microsoft Azure



Different implementations: SaaS

- Software as a service (SaaS)
 - Providing individual software or software packages over the internet
 - Often accessed by the user through a web browser
 - Centrally hosted
 - e.g. Gmail, Netflix, etc.



Cloud computing at IndySCC

- Last three years using a platform called Chameleon cloud 
 - Based on Openstack
 - Similar to cPouta
 - "Bare-metal" hardware
- Actual machines hosted in US across many Universities
 - Based on Openstack
 - Wide range of available and reservable machines
- Anyone can create a user there



Training for IndySCC

- We will use CSC's cPouta service 
 - <https://research.csc.fi/-/cpouta>
 - IaaS
 - No bare-metal nodes available
- Creating your own instance
- Modifying an instance for our needs
- Creating a test cluster





Initializing a computing instance in the cloud

Using cPouta and openstack



How to initialize a computing instance in cPouta

1. Add your public key to the system
2. Reserve a node or multiple nodes
 - Specify how many resources you want (cpus, ram)
3. Choose what underlying operating system/environment you want (Images)
4. Specify the security control for your instance
 - Which IP's it can be accessed from
5. Choose which network it will be a part of
6. Wait for it to spin up and log into it

How to customize your instance?

- Install the needed libraries and dependencies
 - Text editors
 - MPI
 - BLAS
 - Etc..
- Create users and working environments
 - Add the necessary ssh keys for outside users to their SSH environment
 - Initialize new paths (e.g /work or /scratch)
- Add additional nodes?
 - Creating a cluster
 - Creating your own network space in cPouta

Let's get to work

- Instructions in the exercises folder
- Read through the exercises and create your own instance

Glossary for terms in cloud computing

Floating IP

- Public, static IP assigned to an instance for it to receive information through the public internet

Image

- A collection of packages that has files, libraries, binaries and some custom settings that makes a container or a vm ready to launch

Instance

- A single virtual machine or bare-metal node in a cloud environment

Snapshot

- Image created and saved based on your current environment setup. Can exclude specific areas on the computer (e.g. /home)

Volume

- Persistent storage space that remains even if instances are removed. They can be attached to new virtual machines and detached from existing ones



Questions?



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